Fiscal Year:	FY 2006	**Task Last Updated:**	FY 06/12/2006
PI Name:	Rydberg, Bjorn Ph.D.		
Project Title:	DNA Double-Strand Break Induction by HZE Particles. Repair & Consequences in Human & Rodent Cells		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Comments:	PI is deceased, as of December 2011.		
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No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
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Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Pluth, Janice (Lawrence Berkeley National Laboratory)		
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	The experiments performed in this project are designed to test the hypothesis that predictable DNA double-strand breaks (DSBs) cause the major cytogenetic and killing effects of HZE particle exposure. We consider the cytogenetic damage induced by such particles as important for carcinogenesis. We predict that cellular effects of HZE particles can be understood by the initial complexity and spatial distribution of DSBs within the cell. To test this, experiments are performed with both low LET radiation (X-rays), low-energy high LET radiation (29 MeV/n N-ions) and HZE particles (400-1000 MeV/n Fe ions). By comparing theoretically derived spatial distributions and complexity of DSBs in the three cases with experimental endpoints (repairability of DSBs, chromosome damage, cell killing), we are testing the validity of such a paradigm for HZE particles. If effects are seen that cannot be explained by the hypothesis, new mechanisms specific for HZE particles would have to be postulated. The project also has an important component in testing bystander effects. Such effects occur in cells that are not directly exposed to radiation, but are exposed to deleterious factors secreted from exposed cells. It has been suggested that bystander effects will dominate at the low doses and dose rates at		

Task Description:	which astronauts will be exposed, since only a small proportion of all cells will experience a hit from a HZE particle, and the remaining cells will be bystanders. In this project we are comparing bystander responses by medium transfer for the three classes of radiation as delineated above. Task Objectives: 1. To measure DNA double-strand break induction by HZE Fe-ions, low energy high LET N-ions and X-rays in the dose-range of 0.1 – 10 Gy to look for specific low-dose effects. 2. To test the possibility that DSBs can be induced in non-irradiated bystander cells. The medium transfer technique is used to test for clastogenic factors in the growth medium of cells irradiated with HZE Fe ions. 3. To measure in human cells the number of DSBs remaining unrejoined 24 hours after irradiation and compare this with remaining unrejoined premature condensed chromosomes (PCC) in the same dose and time interval. 4. To compare micronuclei formation induced by HZE Fe ions of various energies in normal cells and cells defective in non-homologous end-joining. 5. To measure micronuclei formation behind shielding. It is important for risk estimation to verify that there are no unexpected effects from HZE ions compared to other more well-studied high LET ions. Also, mechanistic understanding of bystander effects may help to design countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Low dose radiation data is of general interest for prediction of health effects.
Task Progress:	Progress for the 3rd year of this project relates mainly to Aim 4. We were interested to study a significant biological effect as a function of Fe ion energy and shielding thickness. For this purpose we measured micronuclei (MN) induction rather than clonogenic survival as originally proposed. This assay is much faster and can accommodate many more samples per run, and therefore could still be carried out during the grant period. We measured MN induction for 3 energies of Fe ions (300 MeV/n, 600 MeV/n and 1 GeV/n) for wild type V79 cells and DSB repair-deficient xrs-6 cells. The doses were in the interval of 0.02 Gy to 1 Gy, low enough to be relevant for risk estimations. To obtain the relative biological effectiveness (RBE) for MN induction at the lowest doses possible we fitted second order polynomials to the induction curves and used the initial slopes for RBE calculations. This represents dose values below 0.2 Gy, unusually low for this type of experiments, and therefore of particular significance for risk estimation. For wild type V79 cells, the RBE relative to 300 kVp X-rays was found to increase with increasing LET (up to over 200 keV/µm), while for the DSB repair-deficient xrs6 cells, the RBE did not deviate significantly from 1 regardless of energy. For V79 cells, irradiation with 1 GeV/n Fe ions (LET = 151 keV/µm) gave RBE = 3.1, 600 MeV/n Fe ions (LET = 176 keV/µm) gave RBE = 4.3 and 300 MeV/n Fe ions (LET = 235 keV/µm) gave RBE = 5.7. An RBE equal to one for NHEJ deficient cells is consistent with literature. The data for the repair-proficient V79 cells are interesting, since a considerable proportion of the Fe ion dose will be in the form of delta rays, and only part of the dose will form a track. Furthermore, the track is much wider than a standard high LET radiation such as alpha particle radiation. Both of these considerations would suggest that the peak of RBE vs LET will appear at a higher total LET for an HZE particle than for a low-energy alpha particle. Our experiments thus show a behavior that was expected, supporting the notion that no unexpected phenomenon was present in MN formation from the HZE Fe ions compared with low-energy high LET particles. We also measured MN formation behind plastic shielding material (same material as routinely used at the beam-line to determine the Bragg curve). Various shielding thicknesses were introduced in front of the beam to obtain MN frequency along the whole Bragg curve for 1 GeV/n Fe ions. Special measures were taken to test the efficiency precisely at the Bragg peak where surviving Fe ions stop with very high LET values. This beam undergoes a lot of nuclear fragmentation so the cells were receiving a mixture of ions of lower Z in addition of the Fe ions. The exposure beyond the Bragg peak is exclusively from fragments. We found that MN formation in our cell lines was proportional to the dose profile with the exception at the Bragg peak, suggesting the RBE does not change significantly by introducing shielding materials. At the Bragg peak the efficiency was lower, as expected from an “over-kill” response of cells exposed to Fe ions of very low energy (with LET values above 1000 keV/µm). This experiment suggests that the effect of polyethylene shielding is primarily to lower the dose, while the RBE or quality factor does not change significantly. Our data is presently prepared for publication in peer-reviewed journals.
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